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https://github.com/julianmak/OCES5303_ML_ocean

The repository principally contains the compiled products rather than the source for size reasons.

- ▶ Associated Python code (as Jupyter notebooks mostly) will be held on the same repository. The source data however might be big, so I am going to be naughty and possibly just refer you to where you might get the data if that is the case (e.g. JRA-55 data). I know I should make properly reproducible binders etc., but I didn't...
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OCES 5303 : ML methods in Ocean Sciences

Session 10: FNOs

Outline

- **Fourier Neural Operators**

- the keyword is the **operator** part

- quick recap/overview of **Fourier analysis**

- structure of FNO

- FNO for three examples

- 1d FNO for linear PDE

- 1d FNO for nonlinear PDE

- 2d FNO for image regression

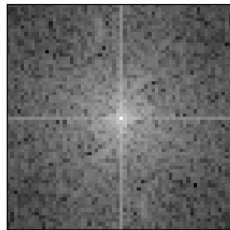


Figure: Head ad-hoc TA and her power spectrum.

Neural Networks vs. Neural Operators

- ▶ **neural networks** are intended to be maps between vectors that live in some **finite** dimensional data space
 - classification of cats/dogs: $(64^2)d$ vector to 1d number
 - dimension reduction of cats/dogs: $(64^2)d$ to 4d say
 - regression of cats/dogs: $(64 \times 32)d$ vector to $(64 \times 32)d$ vector

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- ▶ **operators** map between **infinite** dimension spaces
 - idea: learn the mapping between **spaces**, generalise better to unseen data?

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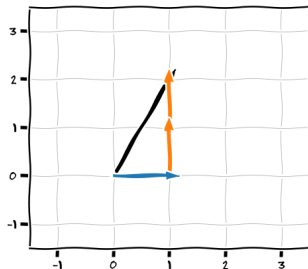
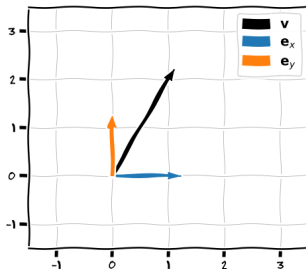
Neural Networks vs. Neural Operators

- ▶ in practice we cannot deal with infinite dimension spaces directly
 - finite ones we can represent as tensors/matrices/arrays
 - idea: take a “good” finite representation of the underlying space, and learn operations acting on that representation
- ▶ **Fourier representation** is one of these
 - basically **sines** and **cosines**
 - can represent a wide class of functions naturally or by artificial modifications
 - others entirely possible (e.g. spherical harmonics)

Recap: Fourier analysis

- ▶ standard basis in $\mathbb{R}^2$ is $e_1 = (1, 0)$ and $e_2 = (0, 1)$, so e.g.

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = e_1 + 2e_2$$

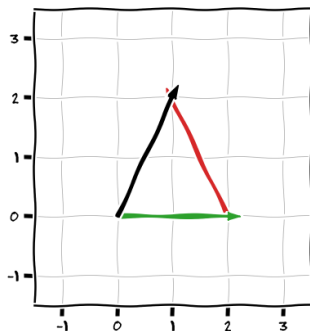
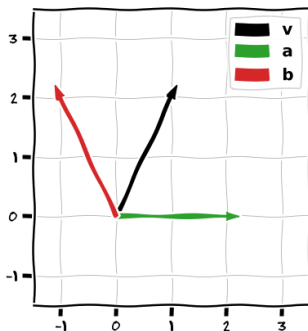


- ▶ $\{e_1, e_2\}$ form a **basis** that happens to be **orthogonal**
(with respect to the usual metric; actually orthonormal here)

Recap: Fourier analysis

- could do instead $\mathbf{a} = (2, 0)$ and $\mathbf{b} = (-1, 2)$, so e.g.

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \mathbf{a} + \mathbf{b}$$



- $\{\mathbf{a}, \mathbf{b}\}$ is still a basis but not orthogonal
→ representation of $\mathbf{v}$ in this basis would be $(1, 1)$

Recap: Fourier analysis

- ▶ idea of a basis is quite general, e.g. $\text{color} = (R, G, B)$
→ cf. lego pieces for the larger structure

Recap: Fourier analysis

- ▶ idea of a basis is quite general, e.g. $\text{color} = (R, G, B)$
→ cf. lego pieces for the larger structure
- ▶ orthogonality is useful because then we have minimal overlaps between the co-ordinates/channels
→ cf. non-confounding variables
→ PCA/EOF looks for an orthogonal representation from data (the O in EOF)

would like an orthogonal basis for functions

Recap: Fourier analysis

- ▶ TL;DR: sines and cosines form an orthogonal basis for a wide class of functions, i.e.

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) = \text{Real} \left(\sum_{k=-\infty}^{\infty} c_k e^{ikx} \right)$$

- ▶ f has a **real** space representation of (for $f(x_i) = f_i$)

$$f(x) = [f_0, f_1, \dots, f_N]$$

and a Fourier/**spectral** representation of (imprecise with $N/2$ below)

$$\hat{f}(k) = [c_0, c_1, \dots, c_{\lfloor N/2 \rfloor}, c_{-\lfloor N/2 \rfloor}, \dots, c_{-1}]$$

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- ▶ really for L^2 periodic functions
→ can manipulate functions to force them to be L^2 periodic

(e.g. padding, tapering)

Recap: Fourier analysis

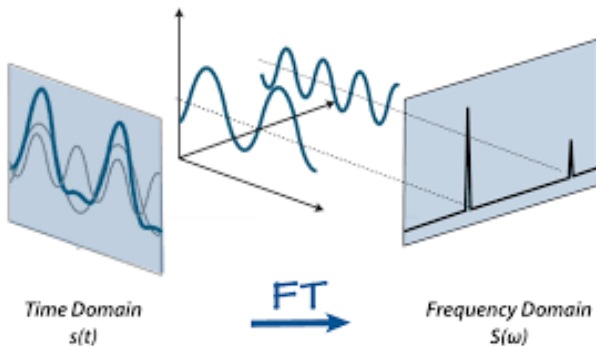


Figure: Schematic for the idea behind Fourier or spectral representation. From the internet (not sure where to attribute...)

- **Fourier transform** from real to spectral representation
→ e.g. signal analysis, imaging, etc.

Recap: Fourier analysis

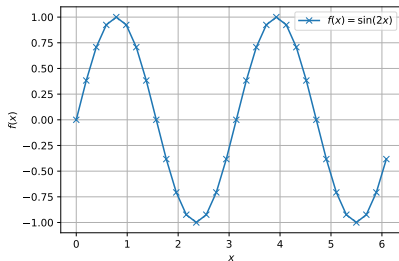


Figure: Example sine wave.

- take domain to be $[0, 2\pi]$ and $f(x) = \sin(2x)$
- Q. what should the spectral representation look like?

$$f(x) = a_0 + \sum_{k=1}^{\infty} a_k \cos(kx) + b_k \sin(kx) = \text{Real} \left(\sum_{k=-\infty}^{\infty} c_k e^{ikx} \right)$$

Recap: Fourier analysis

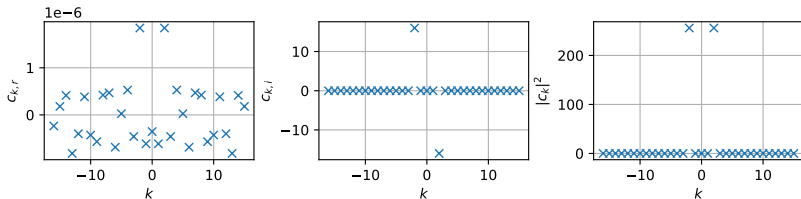


Figure: Fourier transform of the sine wave. (a) Real and (b) imaginary part of the Fourier coefficients, and (c) the power spectrum.

- ▶ done through `torch` (could use `numpy` too)
 - real components of c_k are tiny
 - non-zero imaginary components of c_k only at $k = 2$
 - non-zero **power spectrum** (i.e. $|c_k|^2$) only at $k = 2$

Recap: Fourier analysis

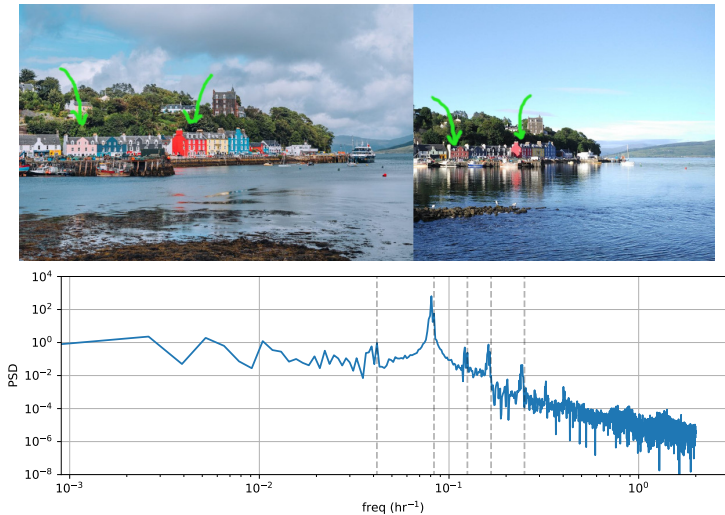


Figure: High and low tide at Tobermory, and its power spectrum computed from tide gauge data.

Recap: Fourier analysis

- ▶ can manipulate Fourier coefficients to do various operations
 - transform to spectral space, manipulate, transform back to real space

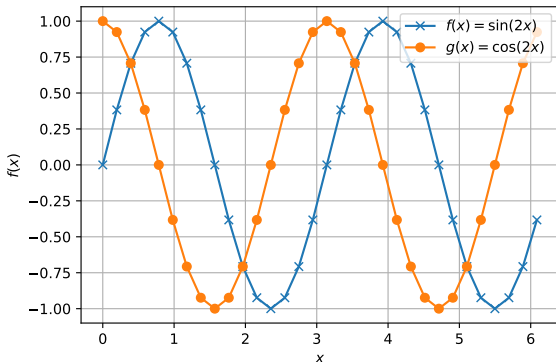


Figure: Example result from manipulating the Fourier coefficients in spectral space: shifting a wave.

Recap: Fourier analysis

- multiply all c_k by ik and $(ik)^2 = -k^2$

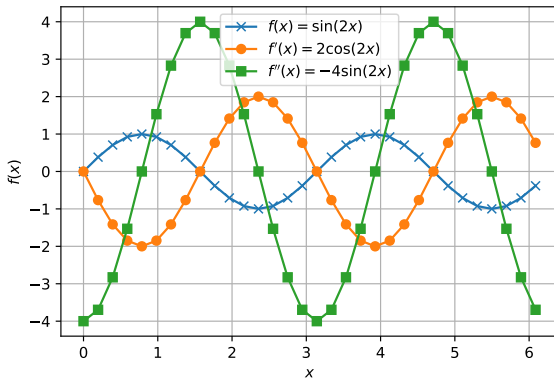


Figure: Example result from manipulating the Fourier coefficients in spectral space: taking derivatives.

Recap: Fourier analysis

- ▶ other manipulations possible, but main one of interest comes from the **spectral convolution theorem**:

$$\begin{array}{c} \text{convolution in one space} \\ \equiv \\ \text{pointwise multiplication in another} \end{array}$$

Recap: Fourier analysis

- ▶ other manipulations possible, but main one of interest comes from the **spectral convolution theorem**:

$$\begin{aligned} &\text{convolution in one space} \\ &\quad \equiv \\ &\text{pointwise multiplication in another} \end{aligned}$$

- ▶ can learn convolutions in real space by learning multiplication in spectral space
 - FNOs like CNNs but with domain-wide kernel
 - can use similar machinery as before, just need to transform the data into spectral space

FNO

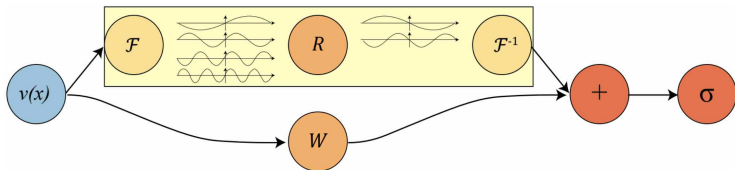


Figure: Schematic for the FNO block. From Fig. 2 of Li et al (2021).

- ▶ **lifting layer** $v(x)$: expands the number of channels
- ▶ **Fourier layer** $\mathcal{F}^{-1}R\mathcal{F}$: transform ($\mathcal{F}$), learn what to multiply in spectral space (R), transform back ($\mathcal{F}^{-1}$)
- ▶ **biases** W : convolution in real space (so multiplication in spectral space) for mode mixing
- ▶ add together $+$, then do activation σ
→ nonlinear activation function to do some **mode mixing**
- ▶ **project** back down at the end

FNO

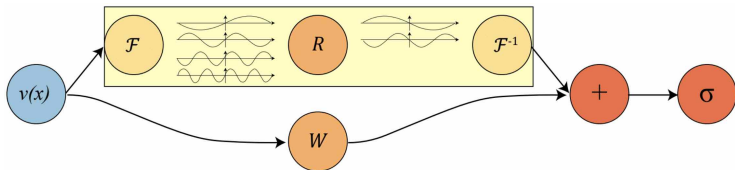


Figure: Schematic for the FNO block. From Fig. 2 of Li et al (2021).

- ▶ grid information concatenated into feature space
- ▶ non-periodicities?
 - can **pad** it (in third example)
- ▶ boundary conditions + nonlinearities?
 - biases and the activation
- ▶ generalisability?
 - should be **resolution** independent

FNO1d: linear PDE

- ▶ do diffusion equation again
 - input is $u(x, t = t_0)$, output is $u(x, t = t_0 + \Delta t)$
 - cf. RNN-type testing

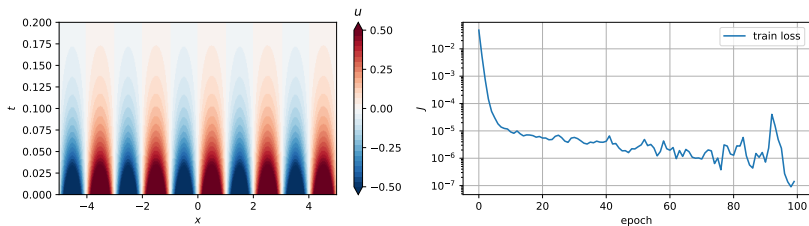


Figure: Data for a particular initialisation and choice of κ from 1d diffusion equation (periodic boundary condition here), and the loss from a simple FNO1d.

FNO1d: linear PDE

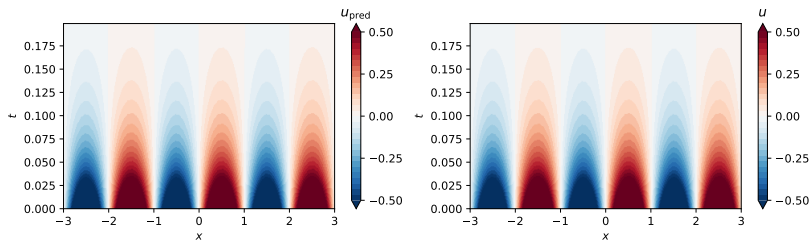


Figure: (Left) One-step prediction: $u_{\text{pred}}(t = t_0 + \Delta t) = F(u(t_0))$. (Right) Model truth.

- ▶ one-step prediction: just predict one time-frame from another
→ pretty good, as it should be, since that's how we trained it in the first place

FNO1d: linear PDE

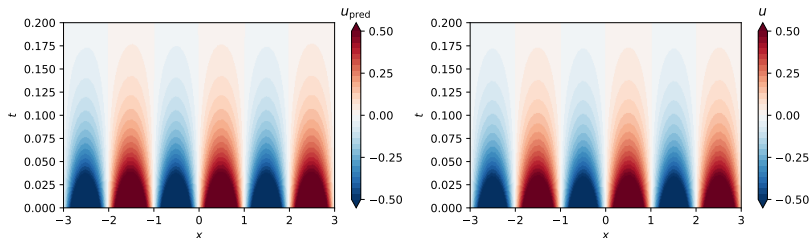


Figure: (Left) Sequential prediction: $u_{\text{pred}}(t = t_0 + n\Delta t) = F^n(u(t_0))$. (Right) Model truth.

- sequential prediction: make one prediction, pass that to model for another prediction and continue
 - much harder test
 - actually seems ok also!

FNO1d: **nonlinear** PDE

► do KdV again

→ input is $u(x, t = t_0)$, output is $u(x, t = t_0 + \Delta t)$

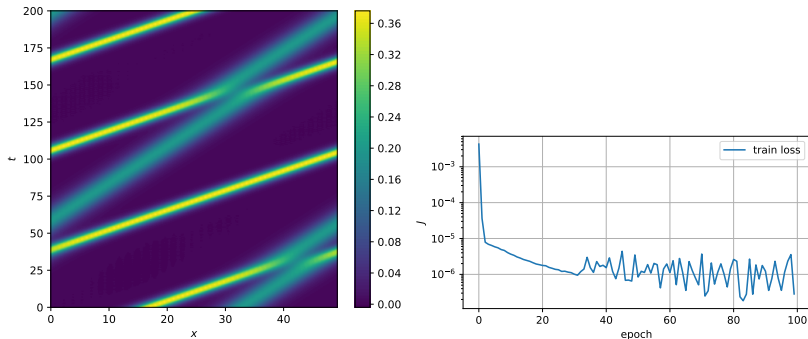


Figure: Data for a particular initialisation from KdV equation (periodic boundary condition here), and the loss from a simple FNO1d.

FNO1d: **nonlinear** PDE

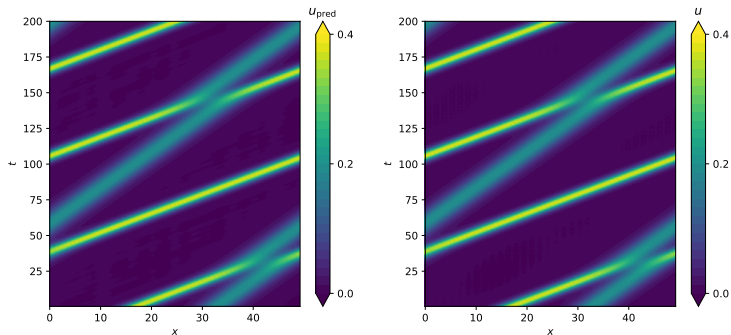


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- one-step prediction: pretty good as it should be

FNO1d: **nonlinear** PDE

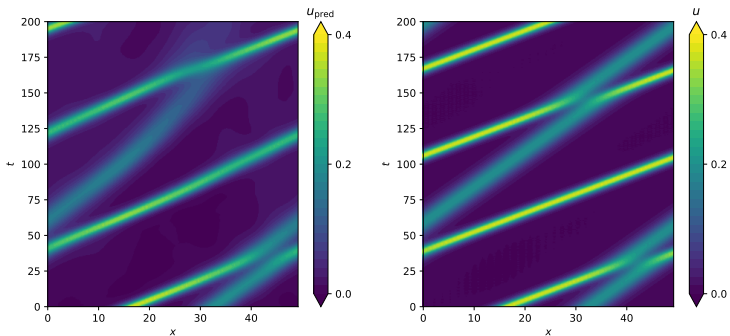


Figure: (Left) Sequential prediction: $u_{\text{pred}}(t = t_0 + n\Delta t) = F^n(u(t_0))$. (Right) Model truth.

- sequential prediction: not so good, but has potential?

FNO2d: image regression

- ▶ FNOs restricted to only differential equations and data with periodic boundary conditions?

FNO2d: image regression

- FNOs restricted to only differential equations and data with periodic boundary conditions?

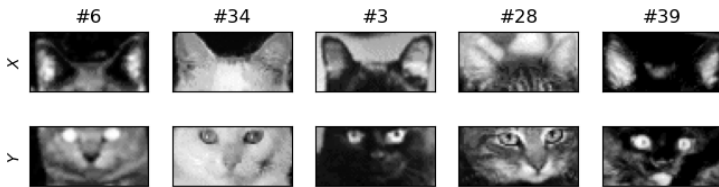


Figure: Unwitting subjects of abuse.

- structure essentially the same, with details as for 2d data
→ adds **padding** to data to formally deal with non-periodic data

FNO2d: image regression

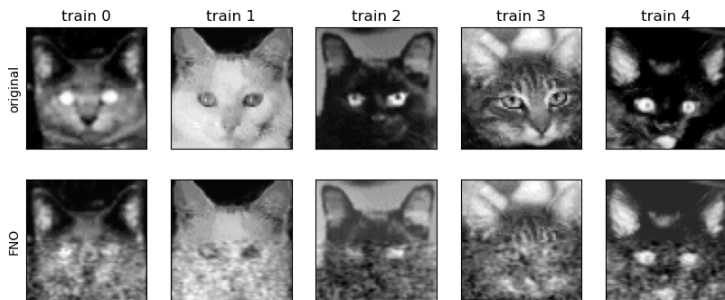


Figure: Unwitting subjects of abuse.

► ...has potential?

FNO2d: image regression

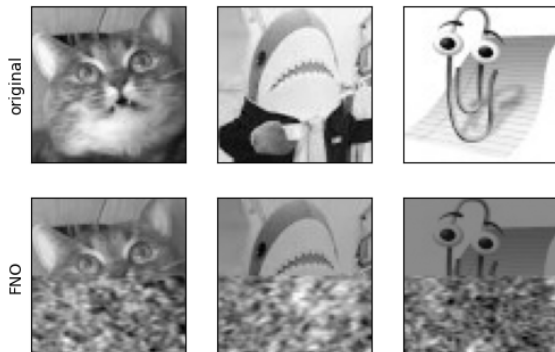
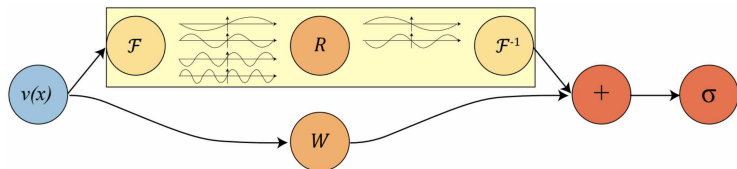


Figure: Further unwitting subjects of abuse.

- ▶ just noise here really
 - I didn't make the model too complex
 - this is a hard problem

Demonstration



- ▶ introduced FNOs
 - main keyword is **operator**, the **Fourier** is one realisation
 - probably most suited to differential equations, but can in principle be used for other things too
- ▶ variants thereof (e.g. RNOs, PINOs, etc.)
 - there is also the `NeuralOperators` package that I didn't use here (but you should try it)

Example variant: PINO

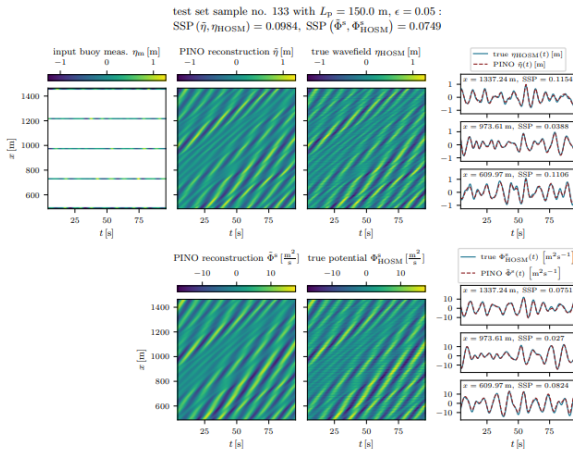


Figure: Reconstructing nonlinear waves with PINO. From Ehlers et al. (2025), Fig. 3.

Moving to a Jupyter notebook →